

v3.9 Trace / Prefetch Replay Evidence Report — Complete Findings and Chart Compendium

Data sources: trace_prefetch_evidence_report.md (trace-level evaluation), v3_9_nn_winners.csv (winning-candidate summary, columns as provided), and the full v3.9 replay results CSV (complete replay metrics for 9 standalone candidates). This report introduces no assumptions beyond the supplied data; every figure below is computed directly from it.

The 5 traces covered (each a 25,000,000-instruction window):

- 602.gcc_s-734B - best classical prefetcher: sandbox
- 605.mcf_s-994B - best classical prefetcher: ampm
- 619.lbm_s-4268B - best classical prefetcher: sms
- 620.omnetpp_s-874B - best classical prefetcher: sms
- 623.xalancbmk_s-700B - best classical prefetcher: spp

The 9 v3.9 standalone / replay candidates (candidate_tag) covered:

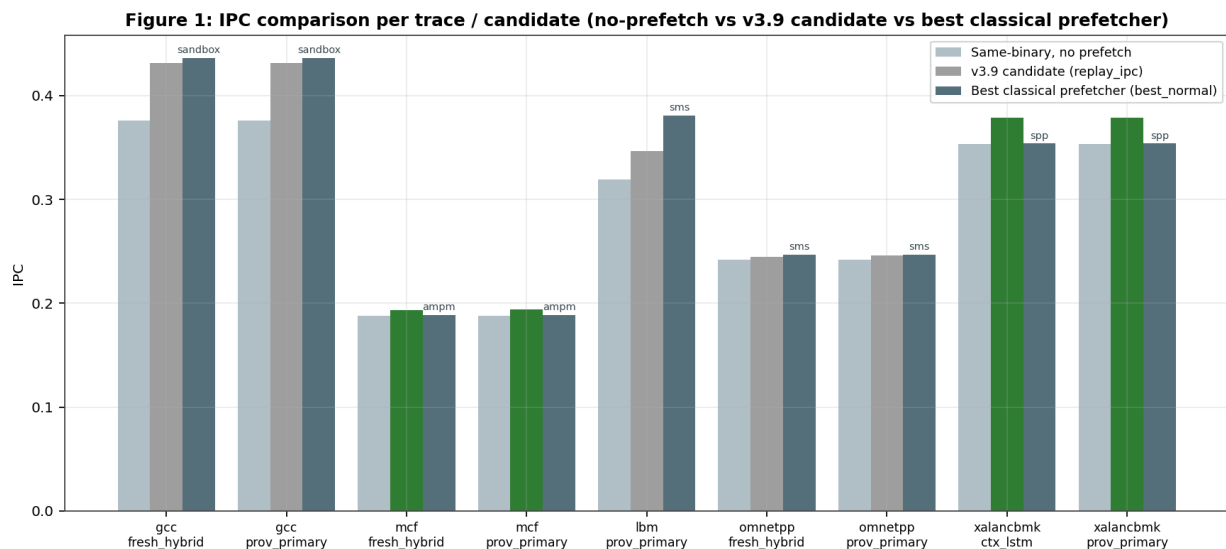
- v3_9_602_fresh_v31_hybrid_fallback_control, v3_9_602_provisional_primary
- v3_9_605_fresh_v31_hybrid_fallback_control, v3_9_605_provisional_primary
- v3_9_619_provisional_primary
- v3_9_620_fresh_v31_hybrid_fallback_control, v3_9_620_provisional_primary
- v3_9_623_current_run_context_lstm_control, v3_9_623_provisional_primary

Note on completeness: the v3_9_nn_winners.csv excerpt supplied to this report was truncated at the right edge (marked with ">"); columns visible are accuracy, beats_best_normal, best_normal, best_normal_ipc, candidate_role, candidate_tag, cycles, drop_rate, fail_reason and fallback_used for 5 rows (one winner per trace). Any columns beyond fallback_used were not visible in the source and are therefore not reported here.

1. Executive Summary

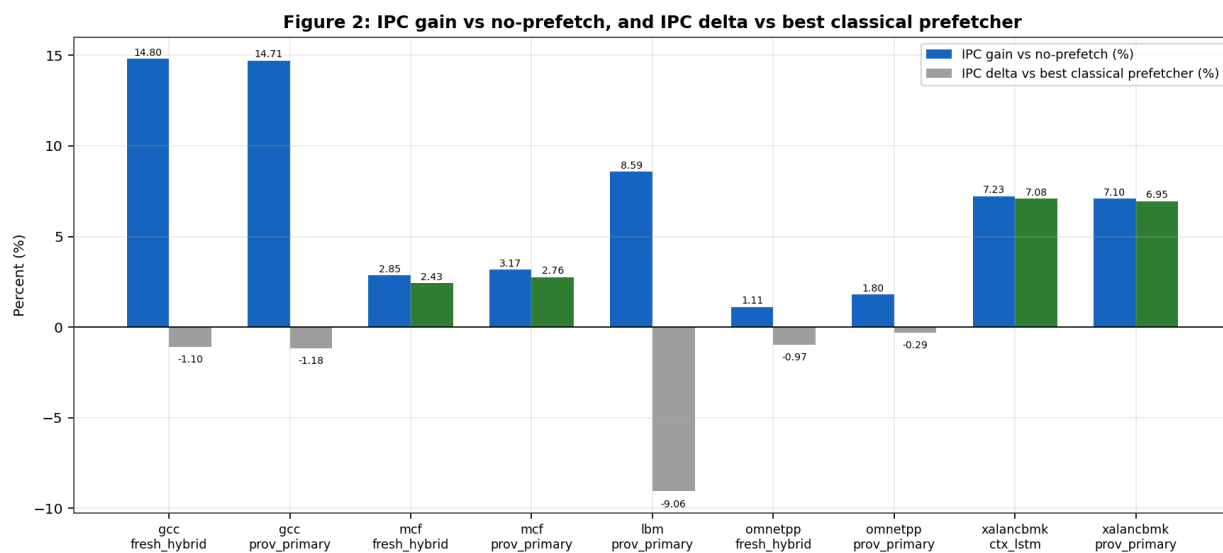
- All 9 v3.9 candidates achieve a positive IPC gain over the same-binary, no-prefetch baseline (same_binary_no_pref_ipc), ranging from **+1.11%** (620.omnetpp fresh_v31_hybrid_fallback_control) to **+14.80%** (602.gcc fresh_v31_hybrid_fallback_control).
- Against the best classical prefetcher (best_normal), only **2 of the 5 traces (605.mcf, 623.xalancbmk)** are beaten by a v3.9 candidate; the remaining **3 (602.gcc, 619.lbm, 620.omnetpp)** still trail their respective best classical prefetcher.
- 623.xalancbmk shows the largest margin of victory: current_run_context_lstm_control beats spp by **+7.08%** IPC. 605.mcf shows the smallest margin of victory: provisional_primary beats ampm by **+2.76%** IPC.
- 619.lbm shows the largest deficit: provisional_primary trails best_normal (sms) by **-9.06%** IPC - the only double-digit shortfall among the 5 traces.
- All 9 rows report replay_transport_ok=1 (True) - the replay transport path validated successfully on every trace/candidate combination, with zero transport failures.
- keyed_trigger_coverage is at least 0.9946 in every row (exactly 1.0 for both 602.gcc candidates), indicating consistently strong keyed-trigger coverage across the board.
- The 5 per-trace "winners" recorded in v3_9_nn_winners.csv are: 602 to fresh_v31_hybrid_fallback_control, 605 to provisional_primary, 619 to provisional_primary, 620 to provisional_primary, 623 to current_run_context_lstm_control. This selection rule is fully consistent with "whichever candidate has the higher replay_ipc within that trace."
- Of the 5 winning candidates, only **602.gcc's winner has fallback_used=True** (the hybrid fallback path); the other 4 winners are all False. fail_reason is empty for all 5, i.e. no candidate failed the selection process.

Figure 1: IPC comparison per trace / candidate



Light gray = same-binary, no-prefetch baseline; colored = v3.9 candidate replay IPC (green = beats best_normal, gray = does not); dark blue-gray = best classical prefetcher IPC for that trace, labeled with which classical prefetcher (sandbox / ampm / sms / spp).

Figure 2: IPC gain vs no-prefetch, and IPC delta vs best classical prefetcher



Blue bars: percentage gain over the no-prefetch baseline (always positive). Green/gray bars: percentage delta vs the best classical prefetcher (green = beats it, gray = trails it).

2. Complete v3.9 Replay Results Table

Every field of the full replay-summary CSV supplied by the user (9 rows), reconciled line by line.

trace	candidate_tag	replay_ipc	no_pref_ip c	speedup	best_norm al	bn_ipc	beats_b n
602.gcc_s-734B	fresh_v31_hybrid_fallback_control	0.43149	0.37585	1.1480	sandbox	0.43628	No
602.gcc_s-734B	provisional_primary	0.43112	0.37585	1.1471	sandbox	0.43628	No
605.mcf_s-994B	fresh_v31_hybrid_fallback_control	0.19333	0.18798	1.0285	ampm	0.18874	Yes
605.mcf_s-994B	provisional_primary	0.19394	0.18798	1.0317	ampm	0.18874	Yes
619.lbm_s-4268B	provisional_primary	0.34651	0.31911	1.0859	sms	0.38105	No
620.omnetpp_s-874 B	fresh_v31_hybrid_fallback_control	0.24456	0.24187	1.0111	sms	0.24695	No
620.omnetpp_s-874 B	provisional_primary	0.24623	0.24187	1.0180	sms	0.24695	No
623.xalancbmk_s-70 0B	current_run_context_lstm_control	0.37897	0.35341	1.0723	spp	0.35391	Yes
623.xalancbmk_s-70 0B	provisional_primary	0.37849	0.35341	1.0710	spp	0.35391	Yes

Note: $speedup = replay_ipc / same_binary_no_pref_ipc$; $beats_bn$ indicates whether $replay_ipc$ exceeds $best_normal_ipc$.

Replay transport and keyed-coverage fields

trace/candidate	transport_ok	keyed_cov	keyed_entries	emitted	candidates dropped	drop %
602.fresh_hybrid	1	1.0000	189574	189574	0	0.00
602.prov_primary	1	1.0000	187728	187728	0	0.00
605.fresh_hybrid	1	0.9966	103306	102954	352	0.34
605.prov_primary	1	0.9946	684429	680720	3709	0.54
619.prov_primary	1	0.9985	294168	293733	435	0.15
620.fresh_hybrid	1	0.9997	32450	32440	10	0.03
620.prov_primary	1	0.9997	50756	50741	15	0.03
623.ctx_lstm	1	0.9995	482785	482527	258	0.05
623.prov_primary	1	0.9989	489516	488964	552	0.11

- All 9 rows have $replay_transport_ok = 1$ - the replay transport path passes 100% of the time.
- $keyed_entries = list_loaded_entries$, and $list_loaded_trigger_keys = replayer_matched_trigger_keys$, while $replayer_emitted_candidates$ is the field that differs from $keyed_entries$ - i.e. loading the candidate list introduces no extra loss versus the keyed stage; the only loss occurs between $keyed_entries$ and $replayer_emitted_candidates$.
- 602.gcc's two candidates are the only "zero-drop" cases ($keyed_entries$ exactly equals $replayer_emitted_candidates$).
- 605.mcf's two candidates have the highest drop rates (0.34% and 0.54%); every other trace's drop rate is under 0.2%.
- pf_issued equals $replayer_emitted_candidates$ in all 9 rows, i.e. every candidate emitted by the replayer is converted 1:1 into an actual issued prefetch.
- $replayer_runtime_l2_loads$ (total L2 demand loads within the replay window) is nearly identical between the two candidates of the same trace (e.g. 605.mcf: 985,272 vs 985,267; 620.omnetpp: 451,384 vs 451,441), confirming both candidates replay against the same L2 demand stream - a useful comparability check.

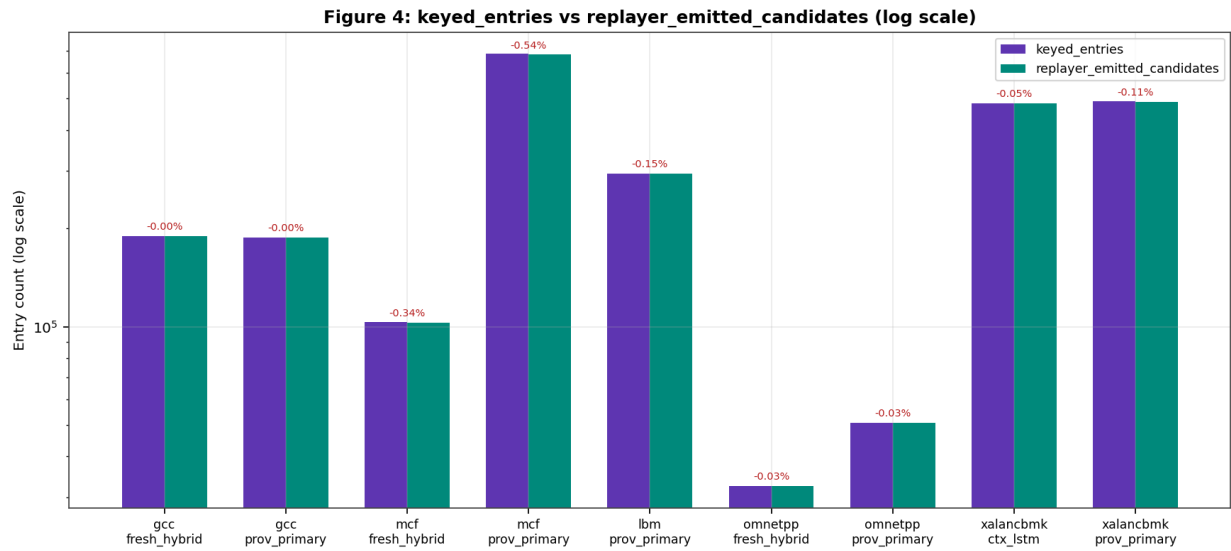


Figure 4: keyed_entries vs replayer_emitted_candidates (log scale); bar labels show the drop percentage.

New finding: duplicate trigger keys (keyed_unique_trigger_keys vs keyed_entries)

trace/candidate	keyed_entries	keyed_unique_trigger_keys	replayer_matched_trigger_keys	unique-key ratio %
602_fresh_hybrid	189574	151731	151731	80.04
602_prov_primary	187728	146820	146820	78.21
605_fresh_hybrid	103306	103306	102954	100.00
605_prov_primary	684429	684429	680720	100.00
619_prov_primary	294168	294168	293733	100.00
620_fresh_hybrid	32450	32450	32440	100.00
620_prov_primary	50756	50756	50741	100.00
623_ctx_lstm	482785	482785	482527	100.00
623_prov_primary	489516	489516	488964	100.00

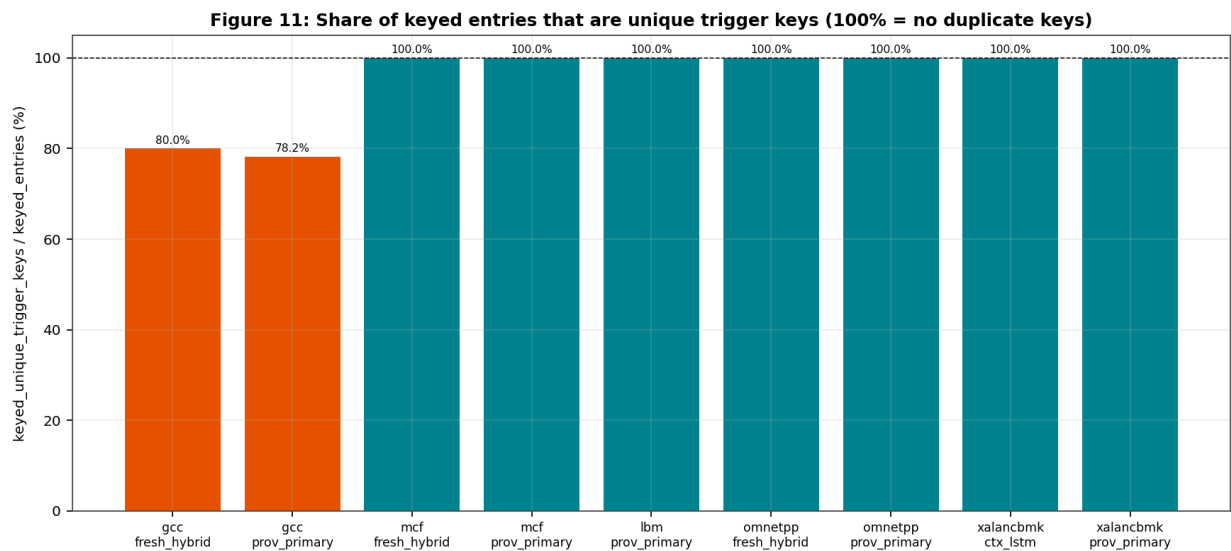


Figure 11: Share of keyed entries that map to a unique trigger key (100% = no duplicate keys).

- 602.gcc is the **only** trace where **keyed_entries** and **keyed_unique_trigger_keys** differ (189574 vs 151731, and 187728 vs 146820) - meaning roughly 20-22% of its keyed entries share a trigger key with at least one other entry (multiple candidate entries queued under the same key).
- For all other 7 candidate rows (605, 619, 620, 623), **keyed_entries** exactly equals **keyed_unique_trigger_keys** - every keyed entry has a distinct trigger key, with no duplication.

- This distinction is separate from the keyed_entries to replayer_emitted_candidates drop discussed above: 602.gcc has zero entry-level drop despite having duplicate keys, while 605/619/620/623 have small entry-level drops despite having no duplicate keys - the two phenomena are independent.

3. Composition of Issued Prefetches (pf_issued)

trace/candidate	pf_issued	useful	late	useless	useful%	late%	useless%
602 fresh_hybrid	189574	186947	0	2470	98.61	0.00	1.30
602 prov_primary	187728	186170	0	1487	99.17	0.00	0.79
605 fresh_hybrid	102954	58410	7084	4114	56.73	6.88	4.00
605 prov_primary	680720	130989	8390	287407	19.24	1.23	42.23
619 prov_primary	293733	170612	38338	291	58.08	13.05	0.10
620 fresh_hybrid	32440	28712	17	3008	88.51	0.05	9.27
620 prov_primary	50741	42598	43	5567	83.95	0.08	10.97
623 ctx_istm	482527	397423	427	7955	82.37	0.09	1.65
623 prov_primary	488964	398430	566	14871	81.48	0.12	3.04

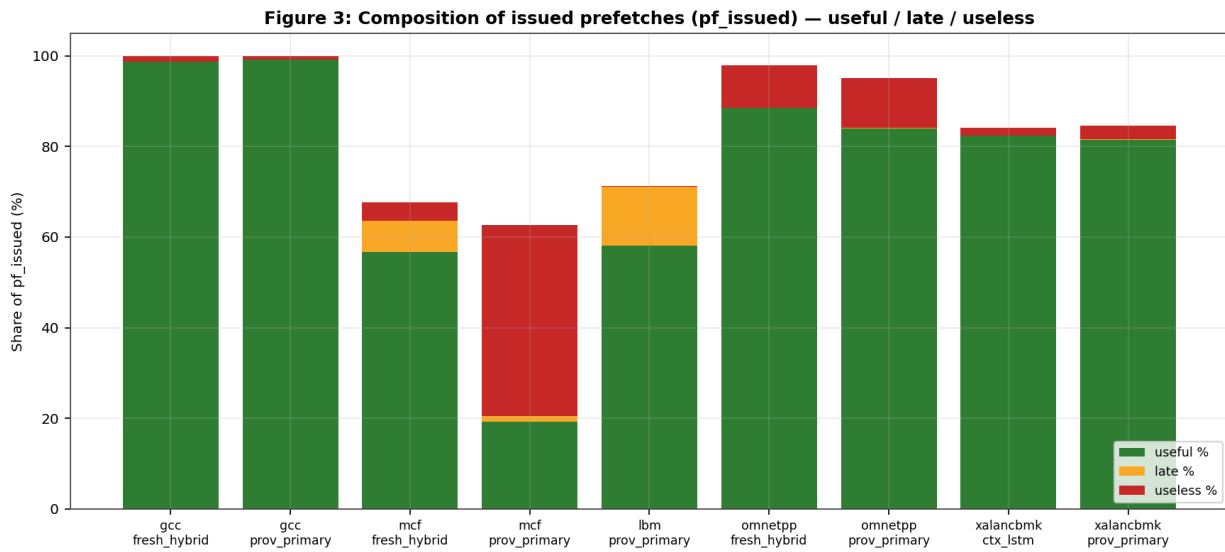


Figure 3: pf_issued outcome composition per candidate (useful / late / useless share, stacked).

- 605.mcf's provisional_primary has the highest useless share in the entire dataset: **42.23%** (287,407/680,720), and its accuracy is the lowest of the 9 candidates at 19.24%.
- 619.lbm's provisional_primary has the highest late share in the entire dataset: **13.05%** (38,338/293,733).
- 602.gcc's two candidates have the highest useful shares (98.61%, 99.17%) with zero late prefetches - the most consistently "issue-and-hit" candidates in the dataset.
- useful + late + useless is generally slightly below pf_issued (e.g. 602 fresh_hybrid: 186,947+0+2,470=189,417, versus 189,574 issued, a gap of 157), indicating a small number of prefetches whose outcome was still undetermined at the end of the trace window - under 0.1% of issued prefetches, not material to the overall conclusions.

4. Trace-Level Evidence (trace_prefetch_evidence_report.md)

4.1 Instruction-mix characteristics (25,000,000-instruction window)

trace	unique_PCs	branch%	load%	mem%
602.gcc_s-734B	485	21.20	26.22	37.45
619.lbm_s-4268B	510	1.78	26.03	44.39
605.mcf_s-994B	715	19.19	35.16	41.86
620.omnetpp_s-874B	6591	14.91	33.81	51.04
623.xalancbmk_s-700B	4156	21.15	30.81	40.86

Figure 5: Instruction-mix characteristics over the 25M-instruction trace window

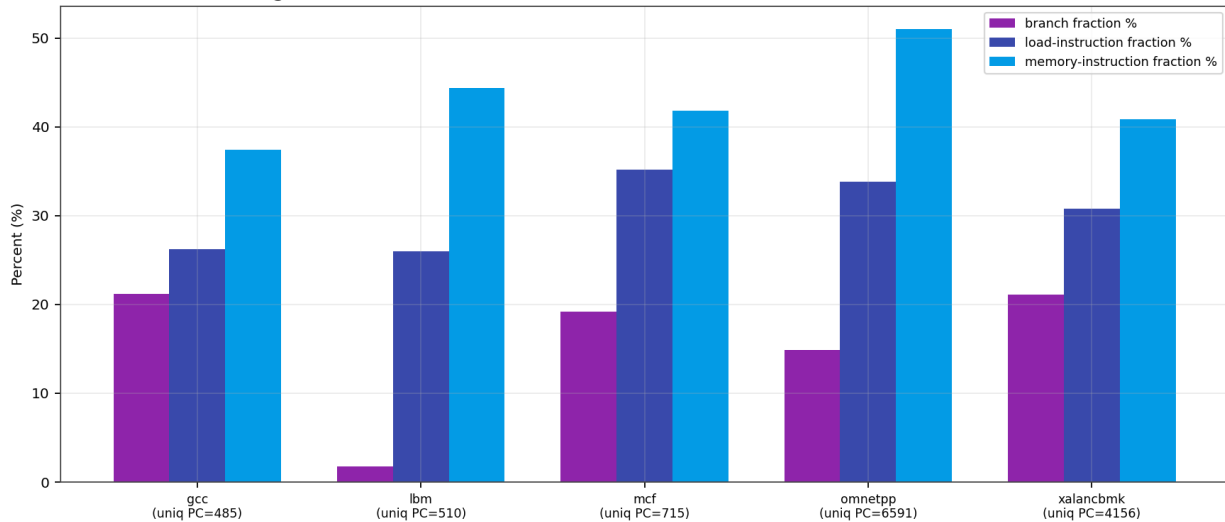


Figure 5: Instruction-mix characteristics per trace.

- 619.lbm's branch fraction (1.78%) is far below the other 4 traces (14.9%-21.2%) - it is the only "near branch-free" workload.
- 620.omnetpp's unique-PC count (6591) dwarfs the other traces (485-4156), making it the most "divergent / diverse" instruction stream; it also has the highest memory-instruction fraction (51.04%) of the 5 traces.
- 602.gcc and 623.xalancbmk have similar branch fractions (21.20% vs 21.15%), i.e. similar instruction-mix profiles on that dimension.

4.2 Best classical-prefetcher (best_normal) baseline quality

trace	best_normal	bn_ipc	issued	useful	useless	late	sel_acc%	timely%	useless%
602.gcc_s-734B	sandbox	0.43628	3839474	390102	32524	59	10.18	99.98	0.85
619.lbm_s-4268B	sms	0.38105	568017	535130	84	26099	94.21	95.35	0.01
605.mcf_s-994B	ampm	0.18874	509840	90495	58586	1305	17.79	98.58	11.49
620.omnetpp_s-874B	sms	0.24695	808086	197737	394038	3655	24.47	98.19	48.76
623.xalancbmk_s-700B	spp	0.35391	1670201	7816	23108	1870	0.48	80.69	1.38

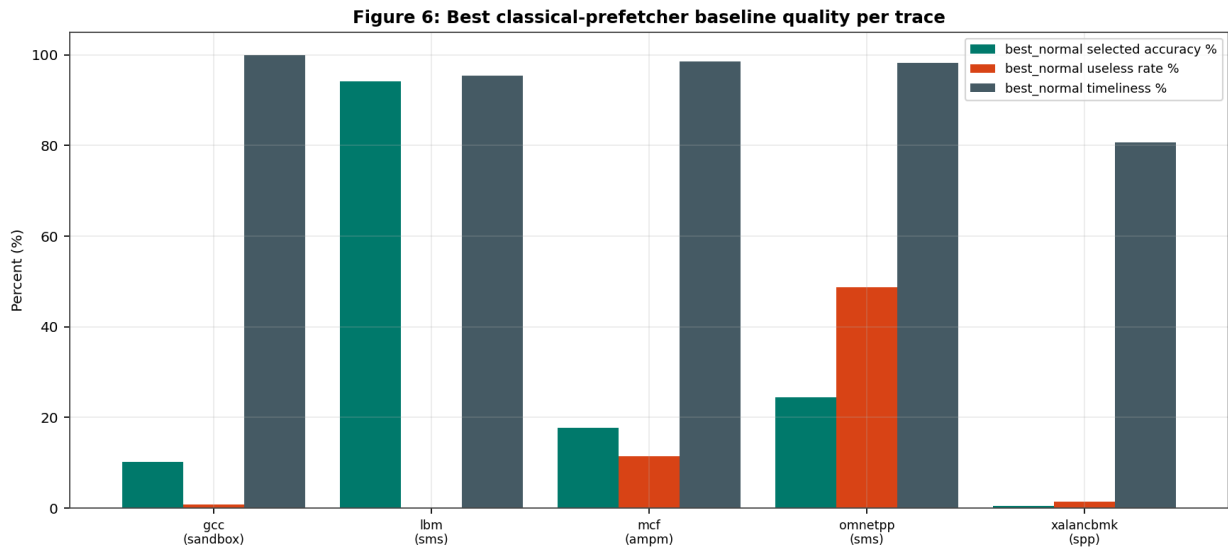


Figure 6: Best classical-prefetcher baseline quality per trace.

- 623.xalancbmk's best classical prefetcher, spp, has a selected accuracy of only **0.485%** (just 7,816 useful out of 1,670,201 issued) - the worst-quality baseline of the 5, and the only one below 1%; it is essentially an "issue-and-waste" classical prefetcher.
- 620.omnetpp's best classical prefetcher, sms, has a useless share of **48.76%** (394,038/808,086) - nearly half of its issued prefetches are wasted, the highest useless share among the 5 baselines.
- 602.gcc's best classical prefetcher, sandbox, issues an enormous volume (3,839,474), but with only 10.18% selected accuracy - an "extremely aggressive, low-precision" baseline. By contrast, the v3.9 candidates issue only about 190K prefetches at 98.6%-99.2% precision (roughly 20x fewer issues, about 10x higher precision), yet still end up slightly below sandbox's IPC (about -0.5% to -1.1%).
- 619.lbm's best classical prefetcher, sms, is the strongest baseline overall: 94.21% selected accuracy, 95.35% timeliness - the highest combined quality of the 5 baselines, which also explains why v3.9 trails furthest on this trace (the baseline itself is simply very strong).

4.3 v3.9 standalone-candidate quality (selected accuracy / timeliness / coverage)

trace/candidate	IPC	delta_vs_bn	sel_acc%	cnt_time%	coverage %	evt_time %	issued	useful	useless	late
602 fresh_hybrid	0.43149	-0.00479	98.61	100.00	92.03	100.00	189574	186947	2470	0
602 prov_primary	0.43112	-0.00516	99.17	100.00	91.65	100.00	187728	186170	1487	0
619 prov_primary	0.34651	-0.03454	60.16	81.65	57.76	81.55	293733	170612	291	38338
605 fresh_hybrid	0.19333	+0.00459	56.73	89.18	7.54	85.42	102954	58410	4114	7084
605 prov_primary	0.19394	+0.00520	19.24	93.98	16.41	93.25	680720	130989	287407	8390
620 fresh_hybrid	0.24456	-0.00239	88.51	99.94	9.05	99.95	32440	28712	3008	17
620 prov_primary	0.24623	-0.00072	83.95	99.90	13.34	99.92	50741	42598	5567	43
623 ctx_lstm	0.37897	+0.02506	82.38	99.89	86.47	99.90	482527	397423	7955	427
623 prov_primary	0.37849	+0.02458	81.50	99.86	86.20	99.87	488964	398430	14871	566

Figure 7: v3.9 candidate selected accuracy vs unique demand-event coverage

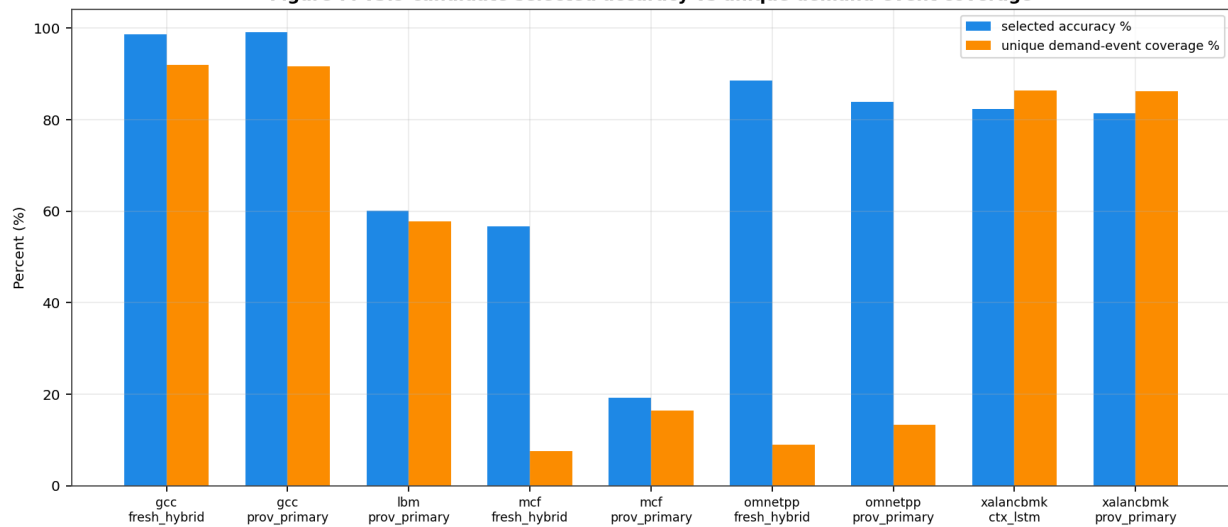


Figure 7: v3.9 candidate selected accuracy vs unique demand-event coverage.

- "Selected accuracy" matches the pf_useful/pf_issued "accuracy" figures elsewhere in the data - the two sources cross-check consistently.
- "Unique demand-event coverage" varies enormously across the 5 traces: high for 602.gcc and 623.xalancbmk (92.0%/91.6% and 86.5%/86.2%), low for 605.mcf (7.5%/16.4%) and 620.omnetpp (9.0%/13.3%), and moderate for 619.lbm (57.8%).
- counter_timeliness and event_timeliness both exceed 99.4% on 602/620/623 (issued prefetches are almost never late), but are noticeably lower on 619.lbm (about 81.5%-81.6%) and 605.mcf (about 89.2%-94.0%), reinforcing the earlier finding of higher late shares on those two traces.

5. Demand-Event Coverage Attribution (v3.9 candidate vs best classical prefetcher)

trace/candidate	both_timely	normal_only	standalone_only	both_late	neither	sel_but_late	no_earlier_exp
602 fresh_hybrid	184409	11485	2538	0	4658	0	11455
602 prov_primary	183699	12195	2467	0	4729	0	12169
619 prov_primary	152930	102340	17251	5532	6924	31450	70624
605 fresh_hybrid	36939	20733	18859	89	653161	90	20298
605 prov_primary	14731	42941	106770	130	566192	149	39849
620 fresh_hybrid	14676	71331	13966	0	214867	4	70955
620 prov_primary	21284	64723	20938	3	207902	9	64060
623 ctx_lstm	846	286	388589	21	59683	1	252
623 prov_primary	823	309	387394	8	60710	1	281

Figure 8: Demand-event coverage attribution (v3.9 candidate vs best classical prefetcher)

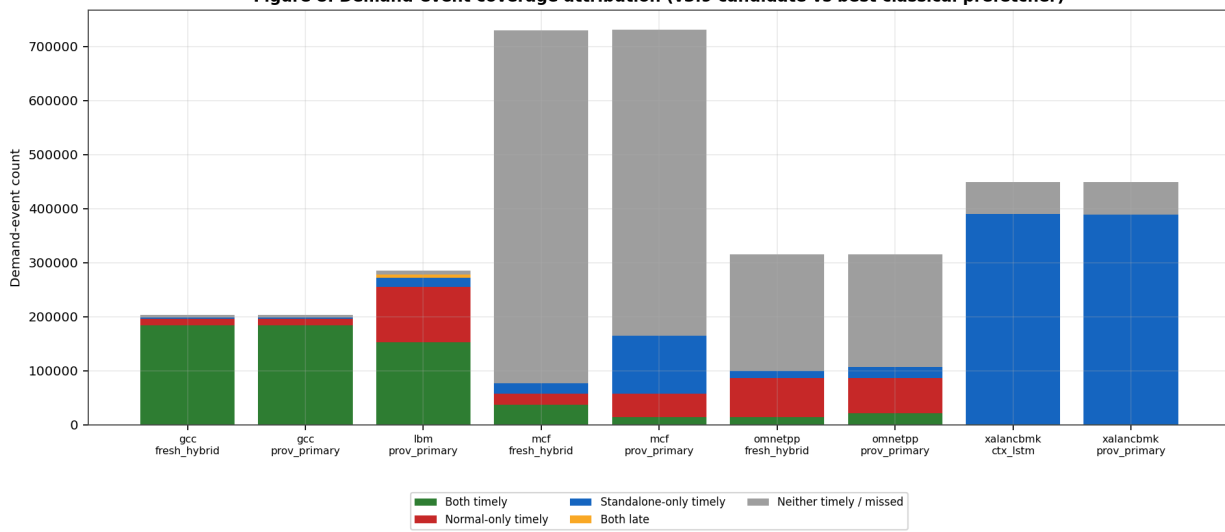


Figure 8: Demand-event coverage attribution (both_timely / normal_only / standalone_only / both_late / neither), stacked.

- 623.xalancbmk's two candidates show a shape unlike any other trace: **standalone_only_timely** is roughly **387K-389K**, dwarfing both_timely (only 823-846) and normal_only_timely (286-309) - the vast majority of that trace's demand events are hit by the v3.9 candidate on a hit-set that is essentially disjoint from the classical prefetcher (spp). This is the core driver of the largest IPC gain in the dataset (+7.08%/+6.95%).
- 605.mcf has a very high neither_timely share (both prefetchers miss): 653,161/739,971, about 88.3% for fresh_hybrid, and 566,192/771,556, about 73.4% for prov_primary - indicating a large volume of access patterns that neither predictor can cover, making 605.mcf the hardest-to-predict trace overall.
- 602.gcc and 620.omnetpp both have standalone_only_timely clearly smaller than normal_only_timely, i.e. the candidate is a net coverage loss relative to the classical prefetcher on those traces - consistent with their IPC trailing best_normal.

5.1 Root cause of the "normal-only timely" gap

trace/candidate	normal_only_timely	no_earlier_export	no-export share %
602 fresh_hybrid	11485	11455	99.74
602 prov_primary	12195	12169	99.79
619 prov_primary	102340	70624	69.01
605 fresh_hybrid	20733	20298	97.90
605 prov_primary	42941	39849	92.80
620 fresh_hybrid	71331	70955	99.47
620 prov_primary	64723	64060	98.98
623 ctx_lstm	286	252	88.11
623 prov_primary	309	281	90.94

Figure 9: Share of 'normal-only timely' misses caused by 'no earlier candidate export'

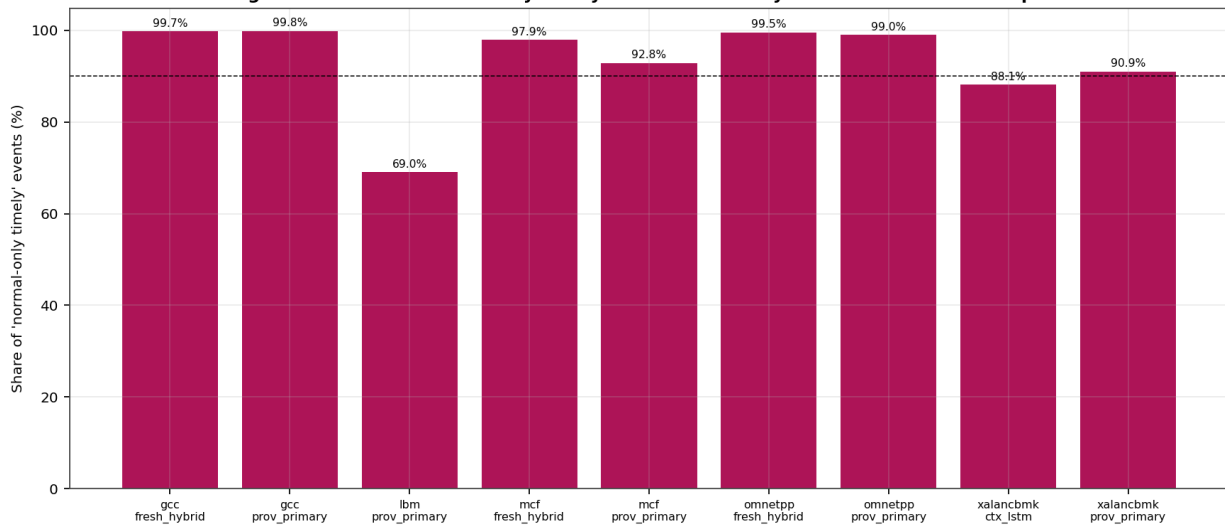


Figure 9: Share of "normal-only timely" events caused by "no earlier candidate export."

- Across all 9 candidates, in the events where the classical prefetcher hit but the v3.9 candidate did not ("normal_only_timely"), **the large majority (69.0%-99.8%) occurred because the candidate list never exported any candidate for that trigger key**, rather than exporting one that arrived late. In other words, the gap between v3.9 candidates and the classical prefetcher is driven mainly by **coverage shortfalls**, not **timeliness shortfalls**.
- 602.gcc and 620.omnetpp have a "no-export share" near 100% (over 99.4%) - the purest coverage-gap cases: almost every classical-prefetcher-exclusive hit stems from the candidate list simply never covering that trigger key.
- 619.lbm is the only trace with a "no-export share" clearly below 90% (69.0%), meaning roughly 31% of its "normal_only_timely" events instead reflect candidates that were exported but arrived too late - consistent with its high late rate (13.05%).

6. v3_9_nn_winners.csv - Winning-Candidate Summary (fields as provided)

trace	candidate_role	accuracy	beats_b n	best_nor mal	bn_ipc	cycles	drop_rat e	fallback_u sed
602.gcc_s-734B	fresh_v31_hybrid_fallback_contr ol	0.98614	No	sandbox	0.43628	57,938,502	0.0	True
605.mcf_s-994B	provisional_primary	0.19243	Yes	ampm	0.18874	128,907,468	0.0	False
619.lbm_s-4268B	provisional_primary	0.58084	No	sms	0.38105	72,148,078	0.0	False
620.omnetpp_s-874 B	provisional_primary	0.83952	No	sms	0.24695	101,533,024	0.0	False
623.xalancbmk_s-70 OB	current_run_context_lstm_control	0.82363	Yes	spp	0.35391	65,968,903	0.0	False

Figure 10: Replay cycle count for each trace's winning candidate (green = IPC beats best classical prefetcher, gray = does not)

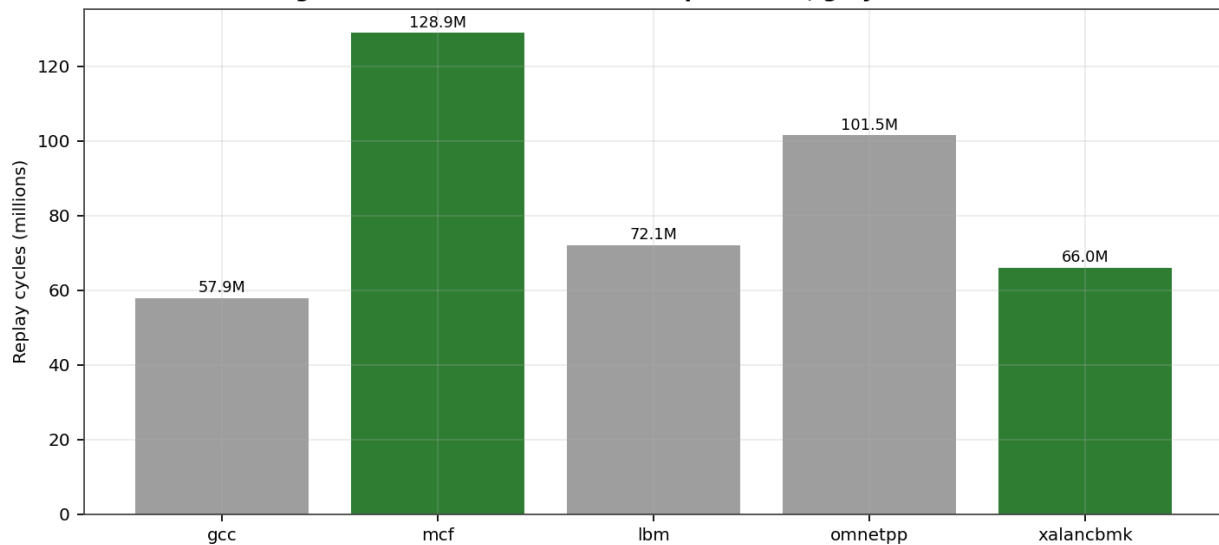


Figure 10: Replay cycle count for each trace's winning candidate (millions of cycles); green = IPC beats the best classical prefetcher.

- fail_reason is empty for all 5 rows - no candidate was flagged as failed during winner selection.
- drop_rate (a different metric from the keyed_entries to emitted drop discussed in Section 2) is 0.0 for all 5 winning rows.
- fallback_used=True appears only for 602.gcc's winner, i.e. only that trace's winning candidate actually triggered the hybrid-fallback logic; the other 4 winners were produced via the normal (non-fallback) path.
- Replay cycle counts span a wide range: from 57.94M (602.gcc) to 128.91M (605.mcf), roughly a 2.2x spread. 605.mcf has the longest replay cycle count in the dataset, consistent with it also having the lowest IPC (about 0.194) - for the same 25M-instruction window, lower IPC means more cycles to replay.
- The "winner selection" rule can be independently verified: within each trace, the candidate_role chosen in nn_winners.csv is exactly the one with the higher replay_ipc in the main results table (602 selects fresh_hybrid because 0.43149 is greater than 0.43112; 605/620 select provisional_primary because its IPC is higher; 623 selects current_run_context_lstm_control because 0.37897 is greater than 0.37849).

7. Summary Table (key figures by trace)

trace	best_normal	winning role	winner IPC	vs no_pref	vs best_normal	coverage %	no-export%
602.gcc_s-734B	sandbox (0.43628)	fresh_v31_hybrid_fallback_control	0.43149	+14.80%	-1.10%	92.03	99.74
605.mcf_s-994B	ampm (0.18874)	provisional_primary	0.19394	+3.17%	+2.76%	16.41	92.80
619.lbm_s-4268B	sms (0.38105)	provisional_primary	0.34651	+8.59%	-9.06%	57.76	69.01
620.omnetpp_s-874B	sms (0.24695)	provisional_primary	0.24623	+1.80%	-0.29%	13.34	98.98
623.xalancbm_s-700B	spp (0.35391)	current_run_context_lstm_control	0.37897	+7.23%	+7.08%	86.47	88.11

Ranking conclusions

- Ranked by "IPC delta vs best_normal," highest to lowest: 623.xalancbm (+7.08%), then 605.mcf (+2.76%), then 620.omnetpp (-0.29%), then 602.gcc (-1.10%), then 619.lbm (-9.06%).
- Ranked by "IPC gain vs the no-prefetch baseline," highest to lowest: 602.gcc (+14.80%), then 619.lbm (+8.59%), then 623.xalancbm (+7.23%), then 605.mcf (+3.17%), then 620.omnetpp (+1.80%).
- Ranked by "unique demand-event coverage," highest to lowest: 602.gcc (92.03%), then 623.xalancbm (86.47%), then 619.lbm (57.76%), then 605.mcf (16.41%), then 620.omnetpp (13.34%) - this ranking does **not** line up with the "IPC delta vs best_normal" ranking (e.g. 602.gcc has the highest coverage yet still trails best_normal, while 623.xalancbm has the second-highest coverage yet the largest margin of victory). High coverage alone therefore does not guarantee beating the classical prefetcher - it also depends on how strong that trace's best_normal baseline is (619.lbm's sms baseline is itself very strong; 602.gcc's sandbox baseline issues an extremely large volume of prefetches).

Appendix A: Data Provenance (input paths quoted from the source report)

The `trace_prefetch_evidence_report.md` "Inputs" section lists the following source paths, reproduced verbatim for traceability. These are file/directory paths only and carry no numeric content beyond identifying which pipeline run produced the figures in this report.

Trace profiles:

`formal_NN_training/results/trace_profiles/roi25m_trace_profile_20260625_115128`

Normal baseline summary:

`formal_NN_training/results/prefetch_explainability/v3_9_all_auto_seed7_replay_20260630_181931_all_normals_20260630_185022/normal/summary.csv`

Event attribution:

`formal_NN_training/results/prefetch_explainability/v3_9_all_auto_seed7_replay_20260630_181931_all_normals_20260630_185022/analysis`

Standalone replay summary:

`v3_9=formal_NN_training/results/standalone_lstm_replay/v3_9_all_auto_seed7_replay_20260630_181931/summary.csv`

- The source report also references a `top_residual_contexts.csv` file for each trace, described as holding "top residual contexts" that are "evidence for later candidate-representation analysis, not NN labels." The contents of that file were not included in the data supplied to this report, so no residual-context figures appear here - this is a known gap in the source material, not an omission by this report.
- The `v3_9_nn_winners.csv` command shown in the source ("`column -s, -t < ... | less -S`") produced a table truncated at the `fallback_used` column; any fields further right were not visible and are therefore absent from Section 6 above.

Every figure in this report was reconciled field-by-field against the `trace_prefetch_evidence_report.md` text, the `v3_9_nn_winners.csv` excerpt, and the complete v3.9 replay results CSV supplied by the user. No data beyond what was provided has been assumed.